



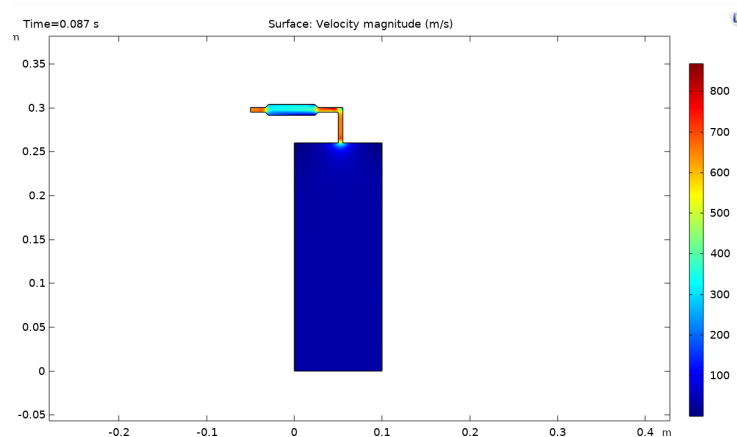
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Sensor calibration

MC/DC Experimental pneumatic set-up

IEM



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Chapter 1: Sensor calibration

1.1 Pressure sensor

The pressure sensor is calibrated against the atmosphere. Thus the measured value when the sensor is exposed to the atmosphere corresponds to 1 bar. The spec sheets indicate a 0.5% error in output, caused by nonlinear, hysteresis and repeatability. Therefore, it is assumed that the zeroing against the atmosphere provides accurate results at 8 bar.

1.1.1 Procedure Pressure sensor

- 1 Expose the sensor to the atmosphere by decoupling and closing valves;
- 2 Connect the sensor to the Arduino/breadboard. Connect black to ground, red to 5v, and green to A0 Arduino. Connect the Arduino to a computer and start Arduino IDE
- 3 Run the calibration code provided below
- 4 Insert calibrated pressureZero and pressureMax in the experimental setup code.
- 5 Connect the sensor to the fire extinguisher pressurized at 8 bar. Measure with compressor pneumatic pistol pressure dial. Make sure it is a closed environment
- 6 Check if the calibration value is correct. Keep in mind that any difference can be caused by multiple things, such as the pressure dial, leaks, incorrect calibration etc.

1.1.2 Calibration code pressure sensor

Download code here: <https://drive.google.com/file/d/1N5TNWYmDKPXWIamPsnneZvXHBETUde2X/view?usp=sharing>

1.2 Flow Sensor

This is a difficult sensor to calibrate without a well-known flow. Here we will approximate the flow using the choking principle. If the pressure ratio between the tank and the outside is two or greater, it can be assumed the flow is choking on the small tube. In these cases, the flow rate is equal to the speed of sound $\approx 343m/s$. This can be supported by a 2D model in Comsol. This shows the following velocities.

The manufacturer states that the calibration factor for water should be 11.

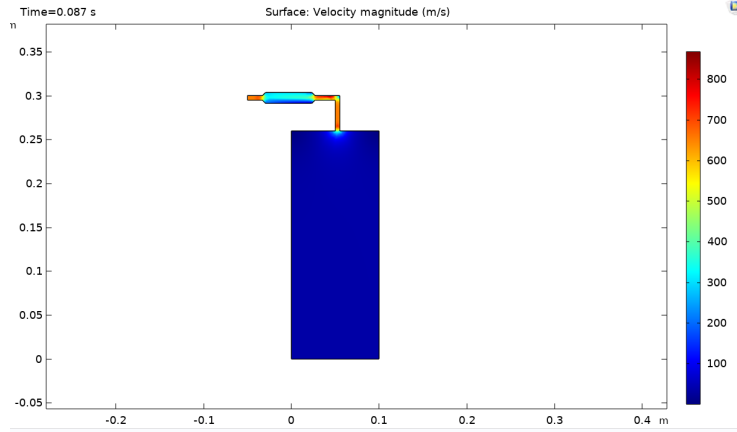


Figure 1.1: Magnitude of velocity in the tank and tubes.

The airflow is defined by:

$$Q_{tube} = Av; \quad (1.1)$$

Where A is the area of the tube: $A = \pi r^2 = \pi(6.25 \cdot 10^{-3})^2 m^2$ and $v = 343 m/s$

- 1 Close the pneumatic system and load it to 8 bars. Connect the sensor to the Arduino and check the pins in the code against reality.
- 2 Start the code, get ready to act quickly. Open the valve quick, when the flow is fully developed start the code and let the pressure run out. Reduce the sampling time if the flow does not stay high long enough.
- 3 Input the calibration factor into the test setup code.

1.2.1 Code Flow

Download code here: <https://drive.google.com/file/d/1NYfv6ezN-nQ1ZO533Tar7QxmUWB7ZQNv/view?usp=sharing>

In this code, the *knownFlowRate* variable is set to your actual known flow rate. The *calculateCalibrationFactor* function uses this known flow rate, along with the tube area and speed of sound, to determine the correct calibration factor. Adjust the values of *knownFlowRate*, *tubeRadius*, and *speedOfSound* based on your specific setup.

1.3 Force sensor

For this calibration, you need the provided weights with masses (provide these masses).

- 1 Securely mount the load cell and make sure it is free from any external loads
- 2 Apply load i to the load cell
- 3 Read and record the output value for the applied load
- 4 Repeat steps 2 and 3 for all loads
- 5 Calculate the calibration factor for the load cell by dividing the known loads by the corresponding load cell output values
- 6 Update the "calibration_factor" value in your Arduino code with the calculated calibration factor

1.3.1 Code Force

Note: Make sure to replace the placeholder comments with actual known loads and corresponding load cell outputs. Additionally, ensure that you're using units consistent with your application (e.g., grams, kilograms, pounds). Consider low loads corresponding with the engine torques to enhance accuracy.

Download code here: <https://drive.google.com/file/d/1Nakx0HwFnjOQpSjgHod1vDwGDWDcIyK0/view?usp=sharing>